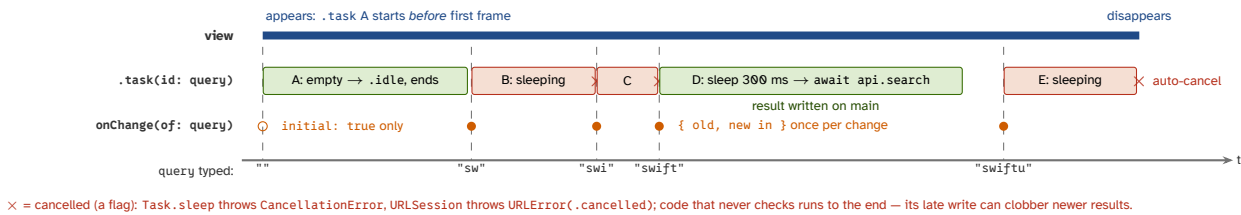


In one line: SwiftUI owns the **lifetime** of async work: `.task` starts with the view and is **cancelled** when the view goes away or its id changes. Views are **@MainActor**: writing state after `await` is safe, blocking the main actor is your bug. Truth lives in `@State` / `@Observable`; effects hang off `.task(id:)`, `onChange`, `.refreshable`.



How it works

- .task { }** (15): starts before the view appears, priority `.userInitiated`; cancelled if the view disappears first.
- .task(id:)**: a new Equatable id value cancels the old task and starts a fresh one. Current docs: created with `Task.immediate` — synchronous up to the first `await`.
- Cancellation is cooperative**: only a flag; `Task.sleep`, `URLSession`, `try Task.checkCancellation()` turn it into an error.
- Isolation**: View is `@MainActor` `@preconcurrency` (Xcode 16 SDK; before, only body). `init`, body, actions, the `.task` closure run on main; `await` suspends without blocking. Swift 6.2 approachable concurrency: a `nonisolated async` func runs on the *caller's* actor; CPU work: `@concurrent`.
- Models**: `@State private var model = Model()` owns an `@Observable`; pass it down as a plain `let`; a child needing `$model.name` declares `@Bindable var model` (or `@Bindable var m = model` in body). Load in `.task`, not `init`.
- onChange(of:initial:_:)** (17): `{ old, new in }` or `{ }`; `initial: true` also fires on appear; main actor, keep it short. `(of:perform:)` deprecated.
- .refreshable** (15): pull-to-refresh; spinner stays **until the closure returns**; `env \.refresh` for custom views.
- Search**: `.searchable(text: $q)` (15) + `.task(id: q)` with a `sleep = debounce and cancel-previous`; `searchScopes`, `searchSuggestions` (16).
- AsyncImage** (15): shared `URLSession`; phases `empty` / `success` / `failure`; no downsampling. **iOS 27** adds HTTP caching, `init(request:)`, `.asyncImageURLSession(_:)`.
- Focus** (15): `@FocusState var f: Field?, .focused($f, equals: .email)`, `.onSubmit { f = .password }`.
- Optional-driven UI**: `.sheet(item:)`, `.alert(_:isPresented:presenting:)`, `now alert(_:item:)`; `nil = dismissed`.

Example — debounced search

```
enum Phase<T> { case idle, loading, loaded(T), failed(String) }
struct SearchView: View {
    @State private var query = ""
    @State private var phase: Phase[Hit] = .idle
    let api: SearchAPI
    var body: some View {
        ResultsList(phase: phase)
        .searchable(text: $query)
        .task(id: query) { // restart per keystroke
            guard !query.isEmpty else { phase = .idle; return }
            do {
                try await Task.sleep(for: .milliseconds(300)) // debounce
                phase = .loading
                phase = .loaded(try await api.search(query)) // on main
            } catch { // incl. URLError
                if !Task.isCancelled { phase = .failed("\(error)") }
            }
        }
        .refreshable { await reload() } // spinner until done
    }
}
```

Which tool for which job

load once per appearance	<code>.task { }</code>
reload when an input changes	<code>.task(id: input)</code>
consume a stream for the view's life	<code>.task { for await v in seq { } }</code>
cheap sync effect on a change	<code>onChange(of:initial:)</code>
user pull / refresh action	<code>.refreshable { await }</code>
button kicks off work	<code>Task { }</code> in the action — lives on; keep the handle to cancel
observe a model outside views	<code>Observations / withObservationTracking</code>

Traps

- !** `.onAppear { Task { await load() } }` — unstructured: **not** cancelled on disappear, outlives the screen.
- !** `try? await Task.sleep(...)` swallows cancellation — the stale “debounced” search still fires. Let it throw, or check `Task.isCancelled`.
- !** Showing `URLError.cancelled` as an error — every keystroke flashes a failure.
- !** JSON decode / image resize inside `.task` runs on the **main actor** — hitches.
- !** `.refreshable { Task { ... } }` returns at once — spinner vanishes.
- !** `.id(UUID())` / unstable identity re-creates the view: `.task` re-runs, duplicate fetches.
- !** The opposite: a split-view detail keeps its identity while selection changes — a plain `.task` never re-runs and shows the old item. Key it: `.task(id: item.id)`.
- !** `onChange` that writes the value it watches (directly or via layout) — a feedback loop; runtime log: “tried to update multiple times per frame” (unverified).

Remember

“The view's life is the task's life; the id is its restart button; main actor by default.”